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Final Project Report

A. Project Overview

Goal: To explore how similarities in students' daily habits (such as study hours, sleep, social media use, etc.) correspond to their academic performance. The project constructs a similarity-based graph to cluster students into behavior-based groups and analyzes performance patterns across these clusters.

Dataset: The dataset contains 1000 rows of student data, including both numeric and categorical fields representing lifestyle factors.

Link to dataset:

<https://www.kaggle.com/datasets/jayaantanaath/student-habits-vs-academic-performance/data>

B. Data Processing

- Loading: The dataset was loaded into Rust using the csv crate (ReaderBuilder). Each row was parsed into a Student struct.
- Transformation:
 - Categorical values such as Yes/No or Fair/Good were encoded as numeric values.
 - Only relevant features were included in the final feature vector.

C. Code Structure

1. Modules

- main.rs: Coordinates execution. Loads data, builds the graph, computes metrics, and prints results.
- graph.rs: Defines the Graph struct and implements graph algorithms (BFS, centrality, clustering, similarity-based connections).
- data.rs: Handles CSV loading and feature vector construction from raw input.

2. Key Functions & Types

- Struct: Student

- Inputs: Row of CSV data
- Outputs: Normalized feature vector and performance score
- Purpose: Represent a student with comparable traits for similarity analysis
- Struct: Graph
 - Purpose: Hold the similarity graph and related algorithms
 - Methods:
 - `build_similarity_graph(k)`: Connects each node to its top-k most similar peers
 - `bfs(start, end)`: Computes shortest path using Breadth-First Search
 - `closeness_centrality(node)`: Computes closeness metric
 - `connected_components()`: Returns list of clusters
 - `component_score_averages()`: Computes average scores per cluster
 - `component_feature_averages()`: Computes average traits per cluster

3. Main Workflow

- `main.rs` loads students → constructs graph → performs BFS → calculates degree and centrality → extracts clusters → summarizes each cluster's score and traits.

D. Tests

```

[~/opt/app-root/src/Final/project2]
● $ cargo test
   Compiling project2 v0.1.0 (/opt/app-root/src/Final/project2)
   Finished _test_ profile [unoptimized + debuginfo] target(s) in 0.75s
   Running unittests src/main.rs (target/debug/deps/project2-4d89c234bb52babb)

running 2 tests
test graph::tests::test_bfs_finds_path ... ok
test graph::tests::test_connected_components_multiple_groups ... ok

test result: ok. 2 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

```

Test 1: `test_bfs_finds_path`

- What it checks: This test builds a small similarity graph of 3 students and ensures that the BFS algorithm can find a valid path between node 0 and node 2.
- Why it matters: Verifies that graph connections are constructed correctly and that `bfs()` traverses the graph as intended, which is essential for pathfinding logic.

Test 2: `test_connected_components_multiple_groups`

- What it checks: This test uses 4 students with distinctly separated features and connects each to only their closest peer ($k=1$). It checks that multiple connected components are formed.
- Why it matters: Confirms that the clustering logic based on similarity (using connected components) works properly and that your graph does not collapse into one giant group — addressing the TA's warning against overly dense graphs.

E. Results

```
[/opt/app-root/src/Final/project2]
$ cargo run
  Compiling project2 v0.1.0 (/opt/app-root/src/Final/project2)
  Finished `dev` profile [unoptimized + debuginfo] target(s) in 2.10s
  Running `target/debug/project2`
Loaded 1000 students.

Graph constructed with 1000 nodes.

Shortest path from 0 to 10 is 10 steps.

Degree distribution:
Degree 3: 1000 students

Closeness centrality of student 0: 0.0858

Found 129 connected components.

Top 5 largest clusters with average scores and features:

Rank 1 : Cluster 1 (800 students)
Avg score: 69.44
age = 20.47
study_hours_per_day = 3.54
social_media_hours = 2.47
netflix_hours = 1.78
part_time_job = 0.21
attendance_percentage = 84.02
sleep_hours = 6.45
diet_quality = 2.20
exercise_frequency = 3.02
parental_education_level = 1.59
internet_quality = 2.31
mental_health_rating = 5.36
extracurricular_participation = 0.30
```

```
Rank 2 : Cluster 5 (10 students)
Avg score: 78.81
age = 21.60
study_hours_per_day = 5.41
social_media_hours = 3.02
netflix_hours = 1.24
part_time_job = 0.40
attendance_percentage = 80.49
sleep_hours = 4.90
diet_quality = 2.30
exercise_frequency = 0.90
parental_education_level = 1.20
internet_quality = 1.90
mental_health_rating = 4.80
extracurricular_participation = 0.20

Rank 3 : Cluster 12 (10 students)
Avg score: 52.95
age = 20.30
study_hours_per_day = 2.15
social_media_hours = 2.69
netflix_hours = 2.81
part_time_job = 0.30
attendance_percentage = 81.38
sleep_hours = 8.03
diet_quality = 2.50
exercise_frequency = 4.80
parental_education_level = 1.50
internet_quality = 2.40
mental_health_rating = 4.50
extracurricular_participation = 0.40

Rank 4 : Cluster 14 (8 students)
Avg score: 52.07
age = 23.00
study_hours_per_day = 2.49
social_media_hours = 3.62
netflix_hours = 2.10
part_time_job = 0.25
attendance_percentage = 88.56
sleep_hours = 7.80
```

```
Rank 4 : Cluster 14 (8 students)
Avg score: 52.07
age = 23.00
study_hours_per_day = 2.49
social_media_hours = 3.62
netflix_hours = 2.10
part_time_job = 0.25
attendance_percentage = 88.56
sleep_hours = 7.80
diet_quality = 2.62
exercise_frequency = 0.75
parental_education_level = 0.75
internet_quality = 2.38
mental_health_rating = 3.25
extracurricular_participation = 0.00

Rank 5 : Cluster 22 (5 students)
Avg score: 79.88
age = 18.00
study_hours_per_day = 4.52
social_media_hours = 3.50
netflix_hours = 2.30
part_time_job = 0.20
attendance_percentage = 76.66
sleep_hours = 5.06
diet_quality = 2.40
exercise_frequency = 4.00
parental_education_level = 1.40
internet_quality = 1.60
mental_health_rating = 9.20
extracurricular_participation = 0.20
```

- The graph was built by connecting each student to their top 3 most similar peers based on lifestyle features (e.g. study hours, sleep, participation).
- A total of 129 clusters (connected components) were identified.
- The largest cluster (800 students) represents a dominant lifestyle group with moderate study, average attendance, and limited outside work.
- This group's average score (69.44) reflects average academic performance.

- Smaller clusters showed more variation in both habits and scores — including high performers with intense study schedules or low performers with low engagement.
- Overall, the analysis shows that similar lifestyle patterns cluster together and correlate meaningfully with academic outcomes.

F. Usage Instructions

- cargo run & cargo test
- No command-line arguments required. Runtime: ~1–2 seconds.

G. AI-Assistance Disclosure and Other Citations

- AI assistance was used via ChatGPT for:
 - Converting CSV values (e.g., encoding categorical variables)
 - Refactoring output formatting
 - Adding interpretation language and modular structure
 - Similarity-based graph construction: I consulted ChatGPT to understand how to connect students based on feature similarity using Euclidean distance, and how to implement k-nearest neighbor logic efficiently in Rust.
 - Graph algorithms: I used AI assistance to help structure the `closeness_centrality()` and `connected_components()` functions using BFS, especially in confirming the correctness of loop logic and traversal behavior.
 - Handling categorical data: For the CSV parsing in `data.rs`, I asked how to convert fields like “yes”/”no” or “poor”/”fair”/ “good” into numeric values usable in feature vectors.